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Cryo-Condensation Technology for Volatile Organic Compound VOC Abatement

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Abstract

The objective of this paper is to address the challenges associated with the Volatile Organic Compound (VOC) emissions during crude oil tanker loading operations. It focuses on applying cryogenic condensation technology to capture and recover VOC, thereby reducing environmental impact and improving operational efficiency. The paper will also reference a case study planned at one of the oil terminals, highlighting the added revenue opportunities through solvent recovery and significant reductions in CO₂ emissions.

The cryogenic condensation module will be introduced as an integrated final polishing step in the hydrocarbon vapors emissions abatement system. The VOC emitted during crude oil tanker loading are adsorbed on carbon beds, then condensed into liquid form; where the recovered products can be reused or sold, creating economic and environmental value. The study will also outline how this modular technology was selected, engineered, and adapted for large marine loading operations. Moreover, key comparisons will be made to conventional methods like thermal oxidation and standard carbon adsorption systems to emphasize the advantages of the cryogenic condensation technology.

The expected results from implementing the cryogenic solution particularly the CIRRUS Vapor Emission Control (VEC) system at the oil terminal indicate VOC emissions reductions of over 95% and converting them into valuable recovered hydrocarbons, avoidance of up to 3,900 tons/year of CO₂ emissions. Initial modeling suggests substantial CO₂ emissions savings by avoiding combustion-based abatement. Additionally, the recovered vapors represent an added revenue stream, improving the return on investment. Observations from similar installations confirm the cryogenic condensation technology's operational reliability, safety (due to inert nitrogen use), and easy integration with marine terminals. These findings underscore the solution's capability to meet increasingly strict environmental regulations while offering economic incentives to operators to reach the highest safety standards. In addition to emission reduction, this paper emphasizes the integration of cryo-condensation system into existing terminal infrastructure, the potential for modular expansion, and its compatibility with emerging decarbonization frameworks. The system's low energy consumption, intrinsic safety, and hydrocarbon recovery capabilities position it as a compelling alternative to traditional VOC abatement strategies.

The novel aspect of this paper is showcasing a scalable, field-ready cryogenic solution to one of the most environmentally challenging phases of oil logistics: the VOC released during tanker loading. By detailing its application at the oil terminal, the paper will provide practical insight into how operators can turn regulatory pressure into a value-added opportunity, contributing both to sustainability goals and production growth. Growing Environmental, Social and Governance (ESG) practices and carbon pricing mechanisms globally are driving renewed attention toward non-combustive VOC abatement strategies, positioning cryogenic condensation systems as enablers of long-term sustainability.

Introduction

Volatile Organic Compound (VOC) are among the most regulated emissions in the oil and gas industry due to their significant contribution to air pollution and human health risks. These emissions can contribute to ground-level ozone formation and photochemical smog, posing both environmental and occupational health hazards. During marine loading operations, displaced hydrocarbon vapors from crude oil tanker loading operations are released into the atmosphere unless abatement technologies are applied.

Regulatory pressures, such as the [U.S. Environmental Protection Agency's \(EPA\) 40 CFR Part 60 Subpart OOOOa](#), requires a reduction of 95% or more in uncontrolled emissions from affected storage vessels. Regional frameworks like the European Union's Industrial Emissions Directive and global initiatives such as the Paris Agreement further drive the adoption of advanced vapor recovery systems.

Traditional technologies such as thermal oxidation, carbon adsorption, and liquid scrubbing have long been employed. However, these methods face drawbacks: high energy consumption, secondary emissions, maintenance complexity, and waste generation. In contrast to cryo-condensation technology, leveraging the ultra-low temperatures of liquid nitrogen (LIN), offers a clean, modular, and energy-efficient alternative, particularly suitable for the intermittent and variable nature of marine vapor streams. The cryo-condensation process cools emissions in a process stream to very low temperatures in innovative heat exchangers. The nitrogen is cycled back into the process for purging and blanketing operations.

Crude oil contains a complex mix of hydrocarbons, including methane, ethane, propane, butanes, pentanes, and aromatics such as BTEX compounds. These components are volatile at atmospheric pressure and temperature, especially during operations like ship loading, tank filling, and barge transfers. As liquid crude oil is loaded onto a vessel, vapor space is displaced, forcing hydrocarbon-rich vapors out of the storage tank or marine vessel compartment. This displacement results in fugitive VOC emissions unless mitigated by a vapor recovery system.

Vapor emissions during marine loading can vary widely depending on the crude oil's vapor pressure, loading temperature, transfer rate, and tank design (e.g., fixed roof versus floating roof). In general, fixed-roof tanks without vapor balancing can emit substantial amounts of VOCs. The concentration levels in these vapor streams can exceed 40 g/Nm³, with flow rates ranging from 100 to over 1,000 Nm³/h. The intermittent and short-duration nature of marine loading cycles adds complexity to emission control efforts, as systems must be capable of ramping up and down rapidly.

The key challenges presented by the marine loading operations include regulatory compliance, economic losses, environmental impact, and operational hazards. Historically, thermal oxidation and carbon adsorption have been deployed to treat marine loading vapors, but they often involve high operational costs, maintenance burdens, or generation of secondary pollutants. With tightening ESG mandates and a focus on sustainability, operators are increasingly seeking technologies that does not only eliminate emissions but also enable the recovery and reuse of hydrocarbons. Therefore, the cryo-condensation modules are introduced to offer dual benefits: emission control and resource recovery. The paper will explore the cryogenic solution, particularly the CIRRUS Vapor Emission Control (VEC) system which provides flexible, compact, and efficient solutions to minimize the atmospheric emissions of Volatile Organic Compounds (VOCs) by using cryogenic liquid nitrogen.

Design Methodology

Increasingly strict air emission regulations challenge the refining industry to tighten control over the release of volatile organic compounds (VOCs). Therefore, cryogenic condensation uses the cooling power of liquid nitrogen to condense the VOCs in a vent or process gas stream. The condensed VOCs can then be captured and either routed back to the process or further treated.

The cryo-condensation leverages the extreme cold of liquid nitrogen (LIN, typically -196°C) to condense volatile organic compounds (VOCs) into liquid form. In the CIRRUS Vapor Emission Control (VEC) system, hydrocarbon-laden vapor is directed through a cryogenic heat exchanger cooled by pre-evaporated liquid nitrogen (LIN). The liquid nitrogen (LIN) is pre-evaporated to generate cold gaseous nitrogen (GAN). Cold (GAN) passes through the thermo-plates of the condenser, cooling down and condensing the VOC which are captured in a tank for re-use or safe disposal. A defrost cycle, using warm nitrogen gas, removes residual ice or solids.

The CIRRUS cryo-condensation systems offer exceptional operational flexibility. They function in batch or continuous modes, suiting facilities with fluctuating vapor flow rates. Modular units (e.g., M50, M150, M500) allow configurations to be tailored to specific capacity needs. Automated switching between operation and defrost modes ensures uninterrupted service and high availability. Integration with Supervisory Control & Data Acquisition (SCADA) systems enables real-time monitoring and diagnostics.

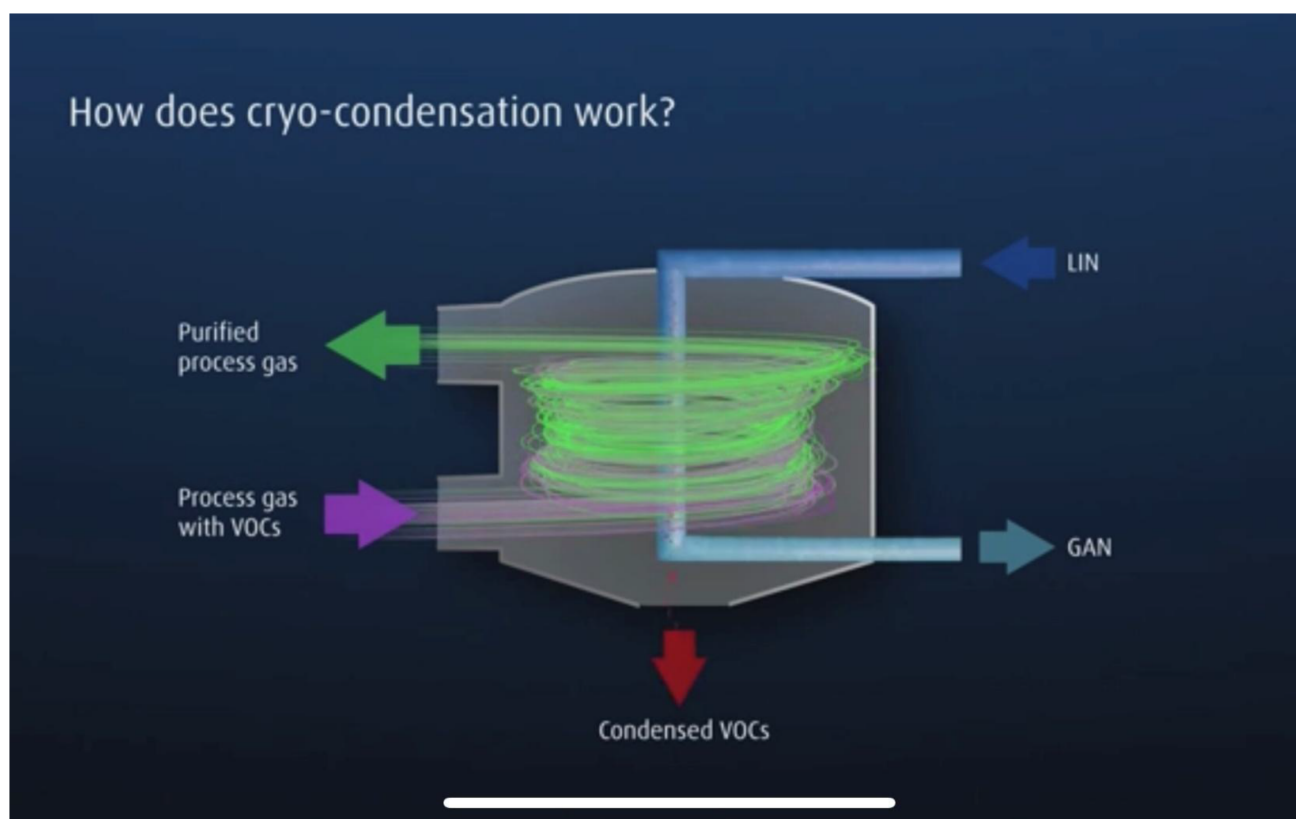


Figure 1—Cryo-condensation functional scheme

Key benefits of technology include:

- High VOC abatement efficiency (>99%), meeting stringent environmental standards
- No secondary waste or emissions, supporting clean operation
- Low energy consumption, as LIN is an efficient cooling agent

- Inherent safety, using inert nitrogen and meeting Atmospheres Explosible (ATEX) requirements
- Hydrocarbons recovery and re-use, enhancing sustainability and reducing operational costs
- Nitrogen does not mix with the VOCs; therefore, both the nitrogen and the condensed VOCs can be re-used for other purposes within the process
- Costs are recovered through nitrogen recycling, hydrocarbons recovery and increased production
- Operators interact with the equipment through a simple and easy-to-use control panel

Field experience and thermodynamic modeling show that cryo-condensation is particularly effective under:

- VOC concentrations from several hundred ppm to percent-level
- Vapor flow rates up to ~40,000 SCFH
- Simple or economically valuable VOC mixtures
- Sites with existing LIN use (e.g., blanketing or purging)
- Environments where combustion or flaring is restricted

Results of Concept

Comparative Analysis: Cryo-condensation vs. Alternatives solutions

Parameters	Cryo-condensation	Thermal Oxidation	Carbon Adsorption
VOC Removal Efficiency	95–99%	95–99%	70–90%
Energy Consumption	Low (LIN only)	High (fuel, electricity)	Moderate
By-products	None	CO ₂ , NO _x	Spent carbon
Product Recovery	Yes (liquid HC)	No	Limited
Maintenance	Low	High	Medium (media change)
Safety	Inert, non-combustive	Combustive	Fire risk from media
Scalability	High (modular)	Limited	Limited

A feasibility study was developed for one of the planned oil terminals with process simulation data and operational measurements to evaluate the cryo-condensation suitability during crude oil tanker loading operations. The VOC sampling could be performed at the vapor header connecting the tank farm and loading arms using a flame ionization detector (FID) calibrated with propane, with data logged every two minutes. Results showed vapor concentrations ranging from 30 to 75 g/Nm³ and flow rates between 18,000 and 27,000 SCFH. Each loading cycle was around 15 hours, with crude transfer rates peaking at 2,500 m³/h. The vapor stream consisted primarily of methane, C₂–C₆ alkanes, and aromatics, typical of high-RVP crudes.

Based on these parameters, the CIRRU M500 cryo-condensation system was selected and configured for batch-mode operation with variable flow capacity. The system was designed for integration downstream of the vapor header, requiring minimal infrastructure modifications. Sensor inputs for temperature, pressure,

and concentration guide LIN consumption and condenser surface optimization. The study confirmed that mobile units could be deployed with flexibility, making cryo-condensation a technical and economical viable solution for high-throughput marine loading terminals, offering both emission control and product recovery without secondary pollution.

The CIRRUS M500 cryo-condensation system is a high-efficiency cryogenic condenser for VOC abatement in hydrocarbon vapor streams. It includes:

- Two plate heat exchangers and a central condenser.
- LIN flows through exchangers to create cold gaseous nitrogen (GAN).
- GAN circulated through the condenser to cool incoming vapor and condense VOCs.

To ensure reliability in high-humidity or moisture-laden environments:

- Dual electric heaters defrost thermo-plates and shell surfaces.
- Heat tracing at the demister outlet prevents ice-related blockages.

The modular platform allows for individual or looped operation (e.g., M150, M500), supporting continuous service even during defrost cycles.

Operating Modes of CIRRUS M500 cryo-condensation system:

- **COOL mode.** The CIRRUS M500 unit needs to be cooled before any VOC is to be condensed. Cooling is achieved by letting nitrogen pass through the unit. The nitrogen enters as liquid nitrogen LIN and then leaves the system as gaseous nitrogen GAN. No process gas should enter the module at this stage to keep the emissions down and to achieve a faster cooling of the unit.
- **RUN mode.** The RUN mode works like the COOL mode (including soft start function) with the exception that when the unit is cooled down, the process gas is allowed to enter the condenser from the bottom. The process gas entering the unit is cooled down to a pre-determined temperature and the VOCs are condensed to liquid. The cooling medium is nitrogen.
- **DEFROST mode:** The CIRRUS M500 unit needs to be defrosted if the process gas is cooled to a temperature close to the freezing point for any of the included VOC components, or if the process gas contains water. The CIRRUS M500 unit uses electrical heaters to heat nitrogen. The unit is equipped with two electrical heaters. The warm gaseous nitrogen then passes through the condenser, and through the double-walled shell; in addition, heat tracing is in the process gas outlet (demister area). The control valve should be open to a fixed position to release the nitrogen.
- **STOP mode:** In this mode, all valves are set to safe mode. No nitrogen or process gas can enter the system. When changing from COOL or RUN mode to STOP mode, the liquid inlet valve opens for 5 minutes and then closes, thus allowing liquid nitrogen to be pushed back into the liquid nitrogen tank.

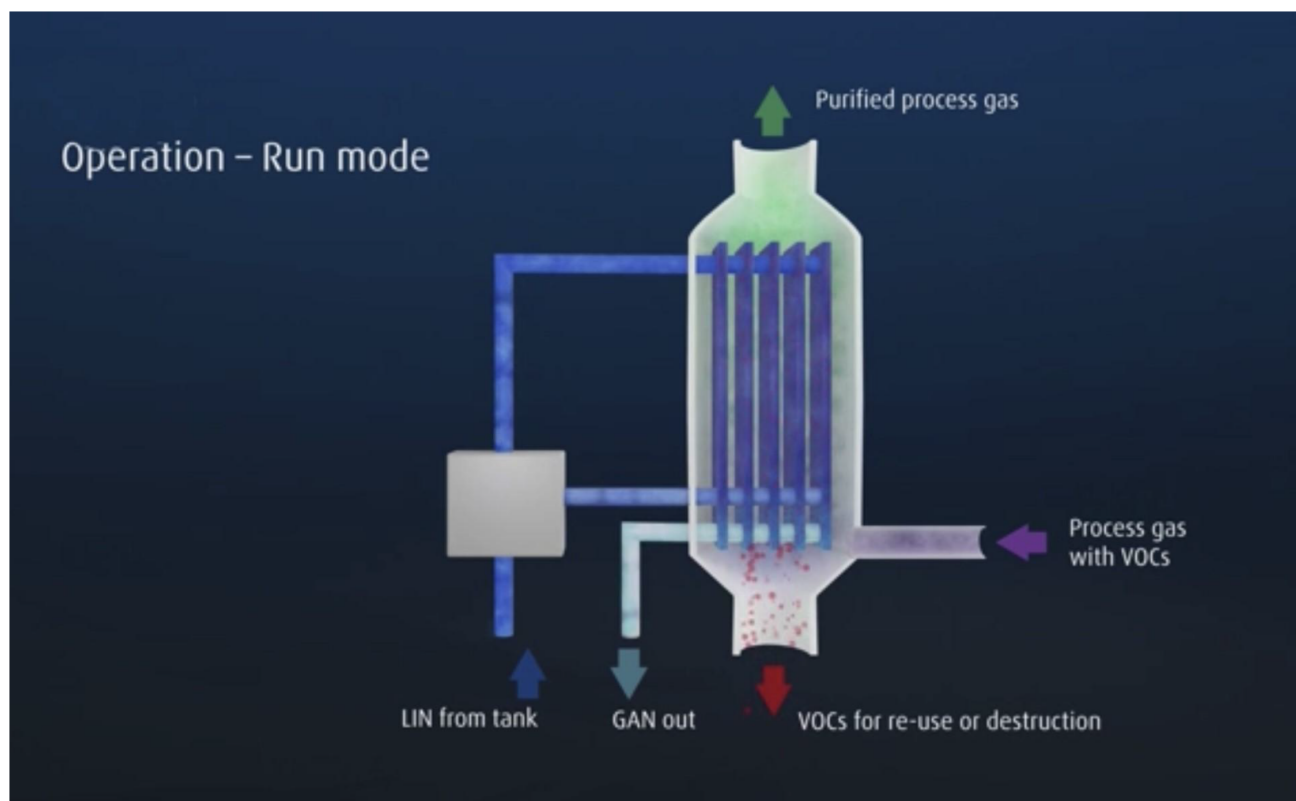


Figure 2—Operation Run mode of the cryo-condensation system

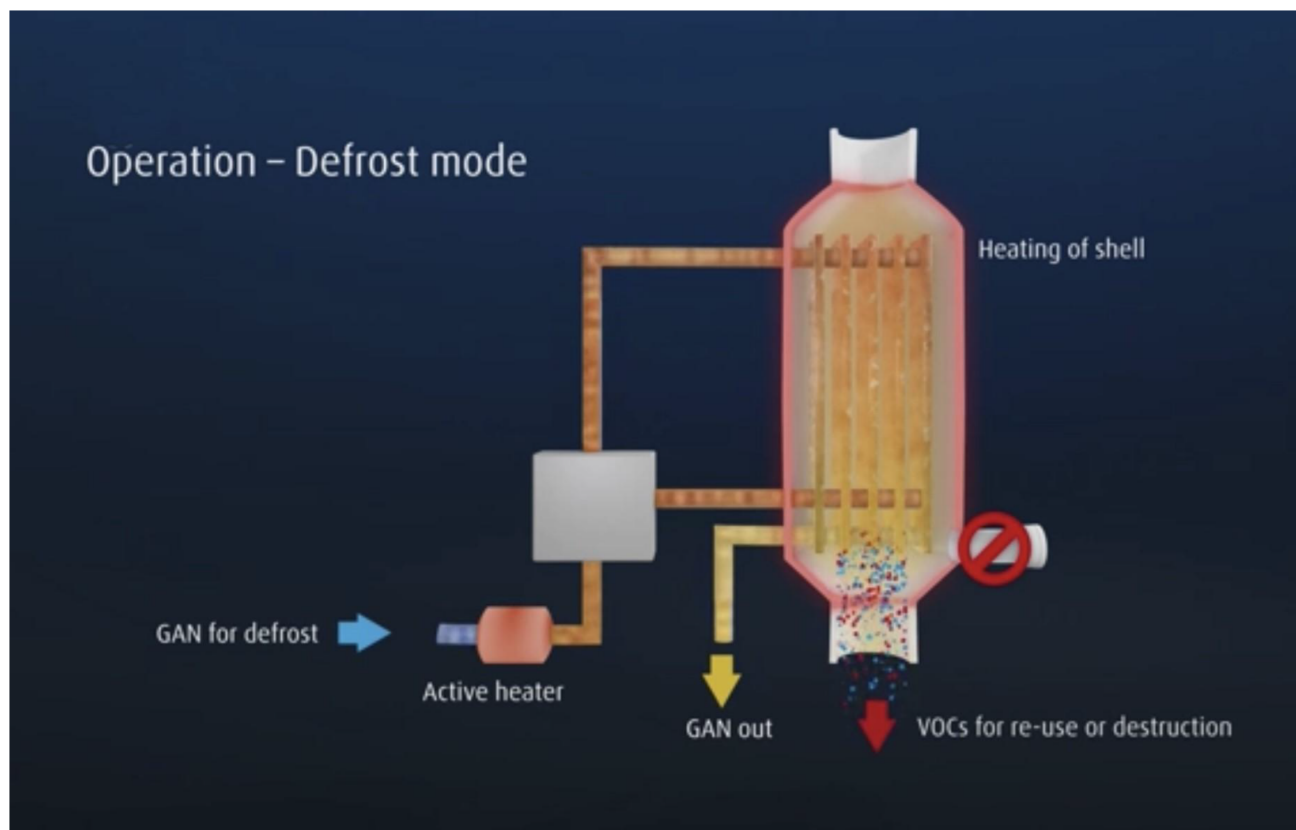


Figure 3—Operation Defrost mode of the cryo-condensation system

Environmental and Economic Summary

Cryo-condensation technology offers significant environmental and economic advantages, particularly relevant to high-throughput facilities like oil terminals. From an environmental standpoint, this solution eliminates combustion-related emissions such as Nitrogen Oxides (NO_x), Sulfur Oxides (SO_x), and Carbon Dioxide (CO₂). As there is no burning process involved, cryo-condensation allows operators to meet stringent environmental regulations without generating secondary pollutants. This supports decarbonization goals and helps meet evolving Environmental, Social, and Governance (ESG) performance frameworks.

Economically, cryo-condensation presents a strong case. Recovered VOCs are condensed into re-usable or sellable liquids, with a market value between \$500 and \$700 per ton depending on hydrocarbon composition and local demand. Capital expenditures for cryo-condensation systems like CIRRUS are lower compared to thermal oxidation systems, as there is no need for burners, scrubbers, or complex waste management infrastructure.

Operating costs are also minimized through nitrogen recycling. CIRRUS units typically consume 30–50% less LIN per ton of VOC removed when compared to conventional cryogenic systems. In jurisdictions enforcing carbon taxes or cap-and-trade mechanisms, the technology enables avoidance of up to \$100 per ton of CO₂ that would otherwise be emitted via thermal destruction.

In practical terms, this translates into a return on investment (ROI) of approximately 12 to 18 months, particularly for terminals with existing LIN infrastructure. The oil terminal, with its established logistics and supply chain for cryogenics, presents a particularly favorable environment for rapid implementation and economic benefit.

Table 1—Economic Summary

Economic Factor	Benefit
Product Recovery	VOCs recovered as reusable or sellable liquids
Market Value of Recovered VOCs	\$500–\$700 per ton, depending on composition
LIN Consumption Efficiency	30–50% lower than traditional cryogenic systems
Avoided Carbon Penalties	~\$100 per ton CO ₂ avoided vs. thermal oxidation
Capital Expenditure	Lower than thermal systems (no burners/scrubbers). The CAPEX of a CIRRUS M500 unit is approx. \$300,000
Operating Costs	Reduced via nitrogen recycling and low maintenance
Return on Investment	12–18 months estimated
Logistics Advantage	LIN supply availability
Operation Cost	The operating costs depend heavily on the actual operating conditions, primarily the VOC load and the inlet temperature. The higher the VOC concentration in the off-gas and the higher the temperature, the greater the cooling demand. As a rule of thumb, 1 kW of cooling power requires 10–12 Nm ³ /h of nitrogen. Therefore, if the maximum cooling capacity of an M500 unit—approximately 80 kW—is fully utilized, the nitrogen demand would be around 800–960 Nm ³ /h. The total operating cost is then calculated by multiplying this consumption by the nitrogen price

This combination of cost efficiency, environmental compliance, and revenue generation from recovered hydrocarbons makes cryo-condensation an ideal VOC abatement solution for oil terminals.

Cryo-condensation systems such as CIRRUS Vapor Emission Control (VEC) offers plug-and-play modular designs that are highly adaptable for both brownfield retrofits and new (greenfield) installations. One of the main advantages observed in real-world deployment is the system's compact footprint, which allows integration into existing terminal layouts without requiring major redesign or overhaul infrastructure.

The flexibility of CIRRUS systems makes them ideal not only for oil terminals, but also for expanding markets such as chemical tank farms, pharmaceutical facilities, bulk solvent storage, and even power plants. These applications often require compliance with stringent VOC emissions threshold regulations that cryo-condensation meets with ease due to its high capture efficiency and inert operating medium.

The integration strategies vary depending on site needs—cryo-condensers can be positioned downstream of adsorption units to polish residual vapors, upstream to reduce thermal oxidizer load, or in parallel as part of a hybrid vapor management scheme. The modularity of the system allows operators to scale up from a single skid to a fully automated multi-unit train with centralized control. This approach is particularly effective for terminals with variable throughput or seasonal demand.

From a digitalization perspective, CIRRUS systems are future-ready. They can be integrated with Supervisory Control And Data Acquisition (SCADA), Distributed Control Systems (DCS), and IoT platforms software to enable real-time performance monitoring. Emerging features such as AI-driven predictive maintenance and cloud-based emissions reporting enhance system reliability while supporting corporate ESG transparency goals. With the push toward Industry 4.0 technologies, these capabilities position cryo-condensation at the forefront of smart terminal operations.

Furthermore, as the industrial sector transitions toward low-carbon operations, the potential use of renewable cryogenics such as green liquid nitrogen generated from wind or solar presents an opportunity to decarbonize even the auxiliary inputs of VOC abatement systems. Several pilot programs are investigating this synergy, including net-zero VOC capture loops powered by clean energy.

In practice, many operators opt for a phased implementation strategy: deploying a mobile or trailer-mounted CIRRUS unit for pilot validation, followed by fixed installation upon confirming performance under site-specific conditions. This modular, scalable, and digitally integrated approach enables both short-term emissions reduction and long-term sustainability.

Key advantages:

- Plug-and-play modular design for brownfield/greenfield integration
- Minimal site modification and compact footprint
- Compatibility with adsorption, oxidation, or hybrid
- IoT software which enabled real-time monitoring and predictive analytics
- Potential for renewable cryogen use (green LIN)
- Scalability for dynamic or expanding terminal operations
- Suitable across sectors: oil & gas, chemical production, pharma/biotech, solvent recovery

Discussion and Conclusion

The CIRRUS Cryogenic condensation technology is a key distinction technique for VOC abatement in refineries & oil terminals. The system's ability to achieve up to 99% VOC recovery while operating with minimal energy and zero secondary emissions makes it remarkably suitable for multiple industry sectors.

The simulated study at one of the planned oil terminals demonstrated technical feasibility, with predicted VOC abatement efficiencies ranging between 95% to 99% under various loading scenarios. Simulations projected vapor flow rates up to 40,000 SCFH, which are well within the handling capacity of the proposed CIRRUS M500 system. Additionally, the modular setup showed that multiple units could operate in parallel to accommodate higher throughput. The economic modeling forecasted a return on investment within 12 to 18 months, driven by hydrocarbon recovery and avoided emissions costs. This supports both regulatory compliance and operational profitability, reinforcing the terminal's potential as a pilot deployment site for full-scale application. The integration into the existing pipelines and loading infrastructure is achievable

without major modifications. The modularity of the CIRRUS M500 system also allows scaling to match terminal throughput.

Future enhancements, including AI-based predictive maintenance and real-time emissions tracking, offer long-term sustainability and automation potential. Given the location and throughput of the oil terminal, the site presents a high-impact opportunity for pilot deployment of the cryo-condensation system.

In conclusion, this technology supports environmental compliance, improves safety, enhances product recovery, and offers rapid ROI which makes cryo-condensation a key contributor to decarbonizing terminal operations and supports the UAE Net Zero emissions by 2045 Strategic Initiative.

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References

- U.S. Environmental Protection Agency (EPA). 2016. Standards of Performance for Crude Oil and Natural Gas Facilities (40 CFR Part 60 Subpart OOOOa). Washington, DC: EPA.
- Eurostat. 2024. EU Emission Trading System Carbon Pricing.
- API. 2014. ANSI/API Standard 2000: Venting Storage Tanks. Linde. 2023. CIRRUS® System Data Sheets. Internal Technical Documentation. Linde. 2020. Solutions to Contain VOC Emissions: A White Paper on Cryogenic Recovery. Linde. 2018. Cryo-Condensation for Control of Storage Tank VOC Emissions. White Paper v2. American Petroleum Institute (API). 2014. *ANSI/API Standard 2000: Venting Atmospheric and Low-Pressure Storage Tanks*. 7th ed. Washington, DC: API.
- U.S. Environmental Protection Agency (EPA). 2003. Air Pollution Control Technology Fact Sheet: U.S. EPA. 2016. 40 CFR Part 60 Subpart OOOOa.
- EPA-452/F-03-021. Research Triangle Park, NC: EPA.